

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1194

Hangman 230 kV
May 02, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1194. The Interconnection Customer proposes a 36.2 MW battery energy storage facility interconnecting at the Hangman 230 kV substation in the Ohio area. The request was received on April 27, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 463 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$459,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BARRETT - MINCO SUBSTATION 230 kV	BARRETT - NORWICH 500 kV	113.0	620	9.6
MISTERSKY - GUADALUPE SUBSTATION 138 kV L	WISSOTA - BERRY ST 138 kV Line	104.4	305	5.1
GUADALUPE - MORTAN 138 kV	BERRY STREET - WISSOTA 138 kV	104.0	324	5.1
MORTAN - MAINWARING 138 kV L	WISSOA - BERRY ST 138 kV	103.9	327	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BARRETT - MINCO SUBSTATION 230 kV	Rebuild line with higher-capacity conductor	\$12,002,000	1% (\$84,000)
MISTERSKY - GUADALUPE SUBSTATION 138 kV L	Reconductor line	\$6,970,000	2% (\$148,000)
GUADALUPE - MORTAN 138 kV	Reconductor line	\$7,719,000	2% (\$164,000)
MORTAN - MAINWARING 138 kV L	Reconductor line	\$2,940,000	2% (\$63,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$459,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.